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DEPARTMENT OF COMPUTER SCIENCE
FACULTY OF PHYSICAL SCIENCES
UNIVERSITY OF BENIN, BENIN CITY

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2024/2025 BSc COMPUTER SCIENCE SECOND SEMESTER EXAMINATIONS
COURSE: CYB203 – CYBERCRIME, LAW AND COUNTERMEASURES
CREDIT UNIT: 03

OPTION B

TIME: 40 minutes

INSTRUCTION: Attempt all questions and shade on your OMR sheet, the option corresponding to the answer choice for each question.

1. One of the most recent legal tools for addressing cyber crimes in the UK is: (a) the Computer Misuse Act 2018 (b) the Data Protection Act 2024 (c) a and b above (d) None of the above
2. What is the primary goal of cybercrime law? (a) To punish hackers (b) To protect individuals and organizations from cyber threats (c) To regulate internet usage (d) To censor online content
3. Which of the following is a type of cybercrime? (a) Identity theft (b) Hacking (c) Phishing (d) All of the above ✓
4. What is the term used to describe unauthorized access to a computer system or network? (a) Hacking (b) Phishing (c) Malware (d) Spyware ✓
5. What is the purpose of encryption in cybersecurity? (a) To block malware (b) To detect hacking attempts (c) To protect sensitive data (d) To speed up internet connections ✓
6. Which of the following is a common method used by cybercriminals to steal sensitive information? (a) Phishing (b) Malware (c) Social engineering (d) All of the above ✓
7. What is the term for malicious software designed to harm or exploit a computer system? (a) Virus (b) Malware (c) Spyware (d) Adware ✓
8. Which of the following is a consequence of cybercrime? (a) Financial loss (b) Reputation damage (c) Legal liability (d) All of the above ✓
9. What is the role of incident response planning in cybersecurity? (a) To prevent cyberattacks (b) To detect cyberattacks (c) To respond to and mitigate cyberattacks (d) To prosecute cybercriminals ✓
10. Which of the following is a best practice for protecting against cybercrime? (a) Using strong passwords (b) Keeping software up-to-date (c) Using antivirus software (d) All of the above ✓
11. A crime is : (a) an act or omission (b) an act (c) an omission (d) an act or conduct ✓
12. All these are characteristics of cybercrime EXCEPT: (a) It is quick, silent and devastating (b) There may be evidence of conflict in the scene of the crime (c) It can be perpetrated in the privacy of one's own home (d) The victim will be aware of the person who committed the crime. ✓

13. The rapid expansion of internet communication technologies and digital activities has created ☒ (a) security concern (b) cyber concern (c) technology concern (d) None of the above
14. The following are examples of cybercrime EXCEPT:
(a) offences against critical national information infrastructure; (b) unlawful access to a computer; ☒ (c) illegal registration of a cybercafé; ☒ (d) non-system interference;
15. Nigeria has the top position in Africa as both the target and origin of harmful cyber operations.
☒ (a) True (b) False (c) a and b (d) None of the above
16. The Nigerian Cybercrime (prohibition, prevention, etc) Act 2015 which addresses cybercrime in the country is similar to: ☒ (a) the UK Computer Misuse Act 1990 (b) the US Cyber Act of 2011 (c) the Ghanaian Cyber crime Act of 1998 (d) all of the above
17. The Nigerian Cybercrime (prohibition, prevention, etc) Act of 2015 does NOT incorporate clauses that pertain to global collaboration and support in the context of cybercrime inquiries and litigations ☒ (a) True ☒ (b) False (c) None of the Above (d) All of the above
18. The Computer Fraud and Abuse Act is a legislative measure in ----- that criminalises a broad spectrum of computer-related transgressions. (a) Nigeria ☒ (b) United States (c) Canada (d) Ghana
19. ----- is charged with the task of investigating cybercrimes in the United States
☒ (a) Federal Bureau of Investigation (b) Federal Institute of Investigation (c) Bureau of Investigation (d) State Bureau of Investigation
20. Collaboration with other jurisdictions can enhance efforts to prevent, detect, and mitigate cyberthreats, which can help Nigeria. This can be achieved through the following EXCEPT:
☒ (a) Public-private partnerships (b) collaboration with government agencies, private sector entities, and civil society organisations. (c) collaborative partnerships ☒ (d) constant visit to study the legal system of other jurisdictions
21. The United Kingdom, like Nigeria, is confronted with a growing number of intricate and highly developed cyber threats encompassing state-sponsored attacks as well as ransomware activities.
☒ (a) True (b) False (c) a and b (d) None of the above
22. The *mens rea* of cybercrime deals with.
☒ (a) intention ☒ (b) actus reus (c) conduct (d) readiness to commit crime
23. ----- is a renowned scholar who proffered different possible definitions of cybercrime
(a) Prof. B.T. Omoregbe ☒ (b) Prof. S.T. Viswanathan (c) Prof. S.T. Viswanathan ☒ (d) None of the Above
24. The following are examples of cybercrimes EXCEPT
(a) Phishing ☒ (b) Cyber Tapping ☒ (c) Hacking (d) Cyber terrorism

25. Cyber Bullying means ----- (a) the same as Cyber stalking (b) the harassment of a person or group using electronic communication (c) a and b above (d) all of the above ✓
26. The following factors contribute to cybercrimes in Nigeria EXCEPT:
 (a) It is rarely reported (b) the growth of digital forensics (c) excessive computer systems (d) Low conviction rates. ✓
27. Cyberstalking often leads to: (a) Online and offline harassment (b) Offline harassment (c) online harassment (d) None of the above ✓
28. The concept of jurisdiction in cyberspace is peculiar to: (a) Nigeria only (b) United States and Nigeria Only (c) United Kingdom and Australia Only (d) None of the above ✓
29. The United States faces an ongoing issue in reconciling the necessity for law enforcement agencies to obtain information with the protection of citizens' privacy rights.
 (a) True (b) False (c) None of the above (d) The problem is peculiar to Nigeria ✓
30. The Legal framework for combating cyber crimes in Nigeria focuses on: (a) Laws only (b) Laws and Institutions only (c) Policy and Laws only (d) All of the above ✓
31. NCWG stands for: (a) Nigeria Cybercrime Working Ground (b) Nigeria Cybercrime Working Group (c) Nigeria Cyber Working Gain (d) Nigeria Crime Working Group ✓
32. ----- serves as the primary legislation governing cybercrimes in Nigeria. (a) the Cybercrimes (Prohibition, Prevention etc.) Act of 2024 (as amended) (b) Economic and Financial Crimes Commission (c) Nigeria Cyber Crimes Act 2015 (d) The Police Act 2004 ✓
33. There is need for further review of the Cyber Security Act, 2024 (as amended) by providing stiffer penalties for those found guilty. (a) I agree (b) I disagree (c) Not Sure (d) None of the above ✓
34. ----- of the EFCC Act charges the Commission with the responsibility of enforcement and due administration of the Act, the investigation of all financial crimes. *providing section 6*
 (a) Section 25 (b) Section 4 (c) Section 15 (d) Section 5 ✓ *no law*
35. The Advanced Fee Fraud and Other Fraud Related Offences Act 2006 provides for the following EXCEPT:
 (a) prohibition and punishment of certain offences pertaining to advanced fee fraud and other fraud related offences (b) combating cybercrimes and other related online frauds. ✓
 (c) General offence of fraud with several ways of committing it (d) stealing of computers.
36. It is on record that Economic and Financial Crimes Commission (EFCC) has not played any significant role in Nigeria's fight against internet fraud:
 (a) True (b) False (c) None of the above (d) a and b ✓



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2024/2025 BSC COMPUTER SCIENCE FIRST SEMESTER EXAMINATIONS
COURSE: INS204 – SYSTEM ANALYSIS AND DESIGN
CREDIT UNIT: 03

TIME: 1 hour

INSTRUCTION: Answer all questions

1. Which of the following best defines a system? A. A random collection of components
B. A group of independent entities C. A set of interrelated components working towards a goal
D. A file structure used for programming E. A user interface
2. Which of these is NOT a common type of system? A. Manual B. Automated C. Hybrid
D. Transparent E. Open
3. Which phase of the SDLC involves determining the needs of the end user? A. Design
B. Implementation C. Requirements analysis D. Testing E. Deployment
4. A major objective of system approach is to: A. Eliminate all human roles
B. Focus only on technical requirements C. View problems holistically
D. Reduce the number of components E. Avoid feedback
5. A non-functional requirement includes: A. User must log in with a password
B. System must support login authentication C. System should respond within 2 seconds
D. Sales data must be stored E. Database should contain 10 tables
6. Which of the following is NOT part of system requirements? A. Functional requirements B. Legal constraints
C. Physical location of the user D. Hardware interface E. Organizational structure
7. A level-0 DFD is also known as: A. Primitive DFD B. Abstract DFD C. Context diagram D. Flowchart
E. Control diagram
8. External evaluation might involve: A. Internal managers B. Programmers C. System users only
D. Independent auditors E. Database designers
9. Which of the following is NOT a benefit typically evaluated? A. Time saving B. User satisfaction
C. Developer interest D. Cost reduction E. Flexibility
10. One of the first steps in system implementation is: A. Launching the marketing campaign B. Creating test data
C. Developing a plan D. Interviewing users again E. Deleting old files
11. A deterministic system is defined by: A. Multiple simultaneous outputs for a single input
B. High dependency on environmental feedback C. Predictable behavior under specified inputs
D. Adaptive real-time learning modules E. Self-regulating feedback loops
12. Which SDLC model is best suited for systems with rapidly changing requirements? A. Waterfall
B. Spiral C. V-model D. Big Bang E. Agile
13. A technical requirement can be differentiated from a system requirement by its: A. Broad organizational focus
B. Abstract goal orientation C. Precise implementation details D. Subjectivity to stakeholder interpretation
E. Legal and regulatory conformance
14. Which phase of SDLC most critically depends on a complete understanding of non-functional requirements? A. Design B. Maintenance C. Analysis D. Testing E. Deployment
15. A feasibility study that assesses whether the organization has the capacity to support a new system is: A. Technical B. Operational C. Economic D. Legal E. Schedule
16. Which of the following cost comparisons best helps in determining Return on Investment (ROI) for a proposed system? A. Development vs. maintenance cost B. Current system operation cost vs.

- new system operation cost C. Vendor vs. in-house development cost D. Licensing vs. subscription cost E. UI design vs. backend optimization cost
17. Feasibility analysis often omits in early-stage decision-making.
A. Legal feasibility ~~B. Emotional resistance from users~~ C. Budgetary constraints ✓
D. Timelines for development E. Technological compatibility
18. Which feasibility type is best evaluated using Monte Carlo simulation or probabilistic forecasting?
A. Operational B. Schedule C. Technical D. Legal ~~E. Economic~~ ✓
19. Which DFD level represents the most granular view of system processes?
A. Level-0 B. Level-1 ~~C. Level-2~~ D. Context diagram E. Entity-Relationship Diagram ✓
20. Which of the following is a direct consequence of incorrect analysis specifications?
~~A. Inconsistent documentation~~ B. Accurate test data C. Optimized file conversion ✓
D. Accelerated system evaluation E. Enhanced network throughput
21. The primary role of a feasibility report is to:
A. Describe coding architecture B. Provide user manuals ~~C. Justify the system investment~~
D. Design ER diagrams E. Test data accuracy ✓
22. Which fact-finding technique yields the least biased data?
A. Interviews B. Questionnaires C. Observation ~~D. Prototyping~~ E. Document analysis ✓
23. Which type of requirement is typically identified after user shadowing? A. Functional
~~B. Operational~~ C. Technical ~~D. Non-functional~~ E. Regulatory ✓
24. Which of the following is a major limitation of document review in fact-finding?
A. High cost B. Subjectivity C. Time inefficiency ~~D. Inaccuracy due to outdated information~~
E. Limited scalability
25. In requirements engineering, traceability primarily helps in: A. User feedback analysis
B. Defining system boundaries C. Ensuring complete documentation ~~D. Linking needs to deliverables~~
E. Training stakeholders ✓
26. What is the most significant challenge in producing technical documentation for system design?
A. Writing in natural language ~~B. Keeping documentation up-to-date~~ C. Visualizing user interfaces ✓
D. Translating pseudocode E. Explaining data types
27. Which system specification task is most dependent on prior feasibility findings?
A. Interface layout B. Logic flow design ~~C. Process modeling~~ ~~D. Hardware platform choice~~
E. Access control configuration ✓
28. Which diagram is least useful for specifying logical process design? A. Flowchart B. Pseudocode
C. state diagram ~~D. ER Diagram~~ E. DFD ✓
29. Which aspect of documentation is most critical for system maintainers? A. Marketing documentation
B. User interface wireframes ~~C. System architecture blueprints~~ D. Help manuals E.
Deployment checklists
30. What ensures consistency between design documentation and implementation? A. Regular code refactoring
B. Version control of diagrams ~~C. Continuous integration~~ ~~D. Traceability matrix~~
E. Data dictionary ✓
31. In system implementation, which factor is most crucial when planning parallel runs? A. Hardware availability
B. User resistance level C. Data redundancy D. System load forecasting ~~E. Cost of dual operation~~ ✓
32. What is the most strategic function of the Education Coordinating Committee during system implementation? A. Approving budget allocations
B. Configuring hardware modules ✓
~~C. Managing user training consistency~~ D. Writing user manuals E. Designing the UI
33. Which of the following best exemplifies a white-box test case during system test data preparation?
~~A. Validating UI flow for errors~~ B. Stress-testing a login form ~~C. Inputting invalid data to check crash behavior~~
D. Verifying loop conditions and branches E. Analyzing black swan scenarios ✓

34. The key reason for post implementation reviews is: A. Budget verification B. Performance benchmarking C. Stakeholder negotiation D. Regulatory auditing E. Data center optimization
35. A poor system test data plan most often leads to A. Project scope creep B. User interface inconsistency C. Functional redundancies D. Latent system faults E. Inadequate documentation
36. Which of the following file conversion challenges is most likely to impact data integrity? A. Index corruption B. Batch job failure C. Operating system mismatch D. Memory leak E. Multithreaded conflict
37. What is the main responsibility of the data control staff during system installation? A. Defining new record structures B. Performing test case analysis C. Monitoring data movement and validity D. Managing file permissions E. Writing software patches
38. During installation, data preparation staff are expected to: A. Program data entry screens B. Configure network protocols C. Reformat and validate data sets D. Review system logs E. Conduct user evaluations
39. The greatest risk of direct installation strategy is: A. Delayed stakeholder approval B. High testing overhead C. Lack of fallback system D. Redundant staffing E. Legal compliance
40. Which approach most effectively mitigates data migration risks during installation? A. Load balancing B. Parallel systems comparison C. Server caching D. Disk mirroring E. Incremental rollout
41. Which of the following would be considered an intangible benefit in system evaluation? A. Reduced server maintenance cost B. Decreased processing time C. Improved customer satisfaction D. Fewer full-time employees E. Lower electricity bills
42. System evaluation that focuses on regulatory compliance falls under: A. Operational analysis B. Cost-benefit assessment C. External audit requirement D. Technical validation E. Strategic review
43. Which evaluation method best assesses system performance under variable load conditions? A. Focus group feedback B. Regression testing C. Stress testing D. Cost-to-benefit ratio E. Risk assessment matrix
44. What is the main purpose of post-implementation recommendations? A. Reduce developer workload B. Suggest training schedules C. Ensure continuous improvement D. Justify project delays E. Identify regulatory violations
45. Which system evaluation metric is most susceptible to subjective bias? A. Net present value B. Usability score C. Throughput rate D. Downtime duration E. Defect density
46. A system that fails to meet an external requirement will likely face: A. Cost escalation B. Increased user training C. Legal consequences D. Storage bottlenecks E. Redundant scripting
47. Which scenario represents a negative net benefit in cost-benefit analysis? A. Training costs outweigh efficiency gains B. Revenue increase after system adoption C. Maintenance costs decrease over time D. ROI greater than 1 E. Break-even in under a year
48. Which of the following best supports quantitative evaluation of systems success? A. Employee morale surveys B. Incident report reviews C. Metrics on processing delays D. Client testimonials E. Stakeholder interviews
49. A benefit realization plan is used in evaluation to: A. Document project risks B. Translate outputs into measurable outcomes C. Allocate task resources D. Align team roles with deliverables E. Approve technical constraints
50. A key limitation of cost-benefit analysis in systems evaluation is: A. High level of automation B. Lack of documentation C. Inability to capture long-term impacts accurately D. Ease of duplication E. Requirement for software modeling



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2024/2025 BSC COMPUTER SCIENCE FIRST SEMESTER EXAMINATIONS

COURSE: IFT212 - COMPUTER ARCHITECTURE

CREDIT UNIT: 03

TIME: 50 MINUTES.

INSTRUCTION: ANSWER ALL QUESTIONS

1. introduced the term computer architecture while working at IBM, and the concept was formalized in what year? A. Intel, 1968 B. Microsoft, 1944 C. IBM, 1964 D. John Von Neumann, 1946 E. None of the above ✓
2. Which of the following components carries control signals from the processor to other hardware components? A. Address Bus B. Control Bus C. Data Bus D. Bus Bus E. None of the above ✓
3. is the duration it takes to access a word from memory once a request has been made. A. Cycle Time B. Fetch Time C. Memory Access Time D. Register Delay E. None of the above ✓
4. is a small high-speed memory location within the processor that holds data and instructions currently in use. A. Cache B. Register C. ROM D. Memory Module E. None of the above ✓
5. What does the acronym RISC stand for in computer architecture? A. Reduced Internal Storage Component B. Reduced Instruction Set Computing C. Read Instruction Sequential Code D. Random Instruction Set Computer E. None of the Above ✓
6. is a subsystem that facilitates and sharing of resources between components within a computer or across multiple systems. A. Interface Unit B. Control Unit C. Computer Box D. Computer Bus E. None of the above ✓
7. Which of the following systems is designed to support high-resolution graphics and specialized computing applications? A. Supercomputer B. Server C. Workstation D. Mainframe E. None of the above ✓
8. What is the process of executing jobs in a group rather than individually called? A. Batch Programming B. Time Sharing C. Batch Processing D. Multitasking E. None of the above ✓
9. is a computing technique that divides tasks into smaller sub-tasks to allow simultaneous execution, thereby improving speed. A. Distributed Computing B. Multi-core Processing C. Time Sharing D. Parallel Computing E. None of the above ✓
10. The is a type of ROM that is erasable only by electrical means. A. EPROM B. DRAM C. EEPROM D. SRAM E. None of the above ✓
11. The architecture uses larger instruction sets and multiple addressing modes to execute complex instructions in fewer lines of code. A. RISC B. MISC C. VLIW D. CISC E. None of the above ✓
12. Older PCs typically used processors operating at speeds of about, while newer PCs use processors operating at about A. 8-bit, 3.7 MB; 64-bit, 4.66 GB B. 16-bit, 3.9 MB; 64-bit, 3.66 GB C. 32-bit, 2.0 MB; 128-bit, 6.1 GB D. 4-bit, 2.5 MB; 32-bit, 3.5 GB E. None of the above ✓
13. A is a single computer system that contains more than one processing unit sharing memory and other system resources. A. Multi-Computers B. Multiprocessor System C. Microcomputer D. Microprocessor E. None of the above ✓
14. The is a register that keeps track of a program's execution and stores the address of the next instruction to be executed. A. Stack Pointer B. Instruction Register C. Base Register D. Program Counter E. All of the above ✓
15. Who proposed the concept of computer architecture for the EDVAC, and in what year was it completed? A. Blaise Pascal, 1946 B. Charles Babbage, 1837 C. John von Neumann, 1950 D. Jacquard Loom, 1968 E. None of the above ✓
16. What is the full meaning of the acronym POST in computer architecture? A. Power-On Security Test B. Processor Output Self-Test C. Power-On Self Test D. Performance Operating System Test E. None of the above ✓
17. The component ensures that all parts of the computer system are working correctly before the operating system is loaded. A. RAM B. BIOS C. Processor D. GPU E. All of the above ✓
18. Which of the following best describes stored concept of the Von Neumann architecture? A. Programs are hardwired into separate control units B. Data and instructions use separate memory modules C. Each instruction is executed in parallel with the next D. Instructions and data are stored together in the same memory E. None of the above ✓
19. What does an opcode specify in a machine language instruction? A. The amount of memory needed B. The operation to be performed C. The address of the data D. The data to be used E. None of the above ✓
20. What is considered the core component of an operating system that manages system resources? A. Kernel B. Scheduler C. Loader D. Compiler E. None of the above ✓

21. In the context of machine instructions, what is the primary role of operands? A. Store the instruction B. Specify the operation ☒ C. Provide data for the operation D. Initiate program execution ☐ E. None of the above
22. Which of the following best differentiates computer architecture from computer model? ☒ A. Architecture focuses on physical design; model focuses on logical design B. Architecture is an abstract platf. model is a specific implementation C. Architecture is hardware; model is software D. Architecture and model are synonymous terms E. All of the above
23. What is the primary purpose of the instruction register in a CPU? A. Stores final results of operations B. Keeps track of memory size ☒ C. Holds the current instruction being executed D. Stores previous user inputs E. All of the above
24. In which generation of computers were high-level programming languages like FORTRAN and COBOL introduced? A. First generation ☒ B. Second generation C. Third generation D. Fourth generation E. None of the above
25. Which of the following best describes how the control unit manages instruction execution in modern CPUs with pipelining? A. It compresses instruction cycles to reduce memory usage B. It performs arithmetic and logic functions in parallel ☒ C. It supervises simultaneous stages of multiple instructions to increase throughput D. It fetches and executes one instruction at a time in strict sequence E. None of the above
26. Which of the following best describes addressing modes in instruction execution? A. They refer to the physical layout of memory cells. B. They are responsible for assigning memory addresses during boot time. ☒ C. They define how an instruction determines the location of its operands. D. They determine how the CPU controls data buses. E. None of the above
27. What is a significant challenge in relation to Von Neumann architecture known as the 'Von Neumann bottleneck'? A. Limited instruction decoding per clock cycle B. Simultaneous access to memory by CPU and GPU. C. Inability of the CPU to execute conditional jumps. ☒ D. Shared bus for data and instructions slows down throughput. E. None of the above
28. The primary advantage of using registers in a CPU is? ☒ A. Faster access speed compared to other memory types B. Higher capacity than RAM C. Greater non-volatility D. Ability to store programs permanently E. All of the above
29. Which error detection technique appends a single bit to a binary sequence? A. Redundancy Code ☒ B. Parity Bit C. Checksum D. CRC E. Kerckhoff Code
30. The acronym ALU stands for: A. Advanced Level Utility B. Arithmetic Linear Unit ☒ C. Arithmetic Logic Unit D. Automated Logic Unit E. Advanced loop unit
31. Which of these components directly controls instruction execution timing? A. Bus Interface B. Memory Unit C. ALU ☒ D. Control Unit E. Registers
32. A computer bus is responsible for? A. Powering the CPU B. Connecting input devices to output ☒ C. Transferring data between components D. Interpreting machine code E. All of the above
33. A 'nibble' consists of? A. 2 bits ☒ B. 4 bits C. 8 bits D. 16 bits E. 32 bits
34. Which of the following is not a valid performance metric in computer systems? A. MIPS B. FLOPS ☒ C. RPM D. CPI E. None of the above
35. The main difference between static and dynamic RAM is? ☒ A. Speed of access B. Whether they are volatile C. Use of transistors vs capacitors D. Use in ROM vs RAM modules E. All of the above
36. The system clock in a CPU is essential for? A. Permanent data storage B. Instruction decoding ☒ C. Synchronizing operations D. Managing I/O requests E. All of the Above
37. The instruction register (IR) stores? A. The final output B. The next memory address ☒ C. The currently executing instruction D. ALU output E. All of the above
38. Which of the following best defines instruction pipelining? ☒ A. Parallel execution of data B. Storing multiple instructions in a queue C. Overlapping instruction phases to improve performance D. Using dual processors E. None of the above
39. The main function of a decoder in the instruction cycle is to: ☒ A. Translate machine code into micro-operations B. Fetch data from RAM C. Execute instructions D. Signal errors E. Overlap instructions.
40. Which of the following does not affect CPU performance? A. Clock speed B. Number of registers ☒ C. Brand name of the manufacturer D. Instruction set complexity E. None of the above

B



DEPARTMENT OF MATHEMATICS
FACULTY OF PHYSICAL SCIENCES
UNIVERSITY OF BENIN, BENIN CITY

W/26 vs 11 E, 12

Option A

SECOND SEMESTER B.Sc. (FULL-TIME) EXAMINATIONS 2024/2025 SESSION

COURSE TITLE: Elementary Differential Equations (MTH202)

TIME ALLOWED: 1 hr 45 mins

INSTRUCTIONS: (i) Write and circle your attendance list serial number on the objective answer paper.
(ii) Attempt all questions by SHADING (using HB pencil) the letter box that corresponds to the correct option.
Information about your Mat. No., Name, Course code, Faculty code and Departmental code must be clearly written and CORRECTLY SHADED. YOU MUST SUBMIT YOUR QUESTION PAPER ALONG WITH YOUR ANSWER SHEET.

- The order and degree of the differential equation $\left(\frac{dy}{dx}\right)^3 + y^2 = 8$ are respectively;
(a) 1, 1 (b) 2, 3 (c) 1, 3 (d) 3, 3 (e) 1, 2
- The order and degree of the differential equation $\sec x = y\sqrt{\frac{dy}{dx}}$ are respectively;
(a) 2, 1 (b) 1, 2 (c) 2, 2 (d) 1, 1 (e) 1, 3
- The order and degree of the differential equation $\left(1 + \frac{dy}{dx}\right)^{\frac{3}{2}} = c \frac{d^2y}{dx^2}$ are respectively
(a) 2, 1 (b) 3, 5 (c) 2, 3 (d) 1, 3 (e) 2, 5
- The order and degree of the differential equation $\ddot{x} + \log(1+t)\dot{x} = 0$ are respectively
(a) 1, 1 (b) 2, 1 (c) 2, 2 (d) 3, 1 (e) 1, 2
- Determine the differential equation whose general solution is $y = cx^4$
(a) $\frac{dy}{dx} = 4x^3$ (b) $x \frac{dy}{dx} = y$ (c) $\frac{dy}{dx} = \frac{4y}{x}$ (d) $\frac{dy}{dx} = \frac{x}{y}$ (e) $x^2 \frac{dy}{dx} = 4y$
- Determine the differential equation whose general solution is $y = ce^{2x}$
(a) $\frac{dy}{dx} + 3y = 0$ (b) $\frac{dy}{dx} - y = 0$ (c) $\frac{dy}{dx} = 3y$ (d) $\frac{dy}{dx} = \frac{y}{3}$ (e) $\frac{dy}{dx} = 4y$
- Determine the differential equation whose general solution is $y = A\cos x + B\sin x$
(a) $y'' + y = 0$ (b) $y' + y = 0$ (c) $y'' - y = 0$ (d) $y' - y = 0$ (e) $y'' + 2y = 0$
- Solve the differential equation $4ydx + xdy = 0$
(a) $yx^3 = c$ (b) $x^4y = c$ (c) $yx = c$ (d) $xy^4 = c$ (e) $xy^2 = c$
- Solve the differential equation $t \frac{dy}{dt} + 5y = 0, y(1) = 1$
(a) $y = t^5$ (b) $y = \frac{5}{t}$ (c) $y = \frac{1}{t^5}$ (d) $y = 5t$ (e) $y = t^2$
- Find the particular solution of the equation $e^x \frac{dy}{dx} = 4$ given that $y = 3$, when $x = 0$
(a) $y = -4e^{-x} + 7$ (b) $y = 4e^{-x} + 7$ (c) $y = 4e^x + 7$ (d) $y = 4e^x + 7$ (e) $y = e^x + 7$
- Solve the differential equation $x \frac{dy}{dx} - 3y = 4x$
(a) $y = Cx^3 - 2x$ (b) $y = Ce^{3x} - 2x$ (c) $y = x + C$ (d) $y = 2x + Cx^3$ (e) $y = -\frac{2}{x} + Cx^3$
- Which of the following is a First Order Homogeneous Ordinary Differential Equation?
(a) $\frac{dy}{dx} = x^2 + y$ (b) $\frac{dy}{dx} = y + x^2$ (c) $\frac{dy}{dx} = xy$ (d) $\frac{dy}{dx} = x + y^2$ (e) $\frac{dy}{dx} = \frac{x+y}{x}$

13. Which of the following is NOT a First Order Homogeneous Ordinary Differential Equation?

(a) $\frac{dy}{dx} = \frac{y}{x}$ (b) $\frac{dy}{dx} = \frac{x+y}{x}$ (c) $\frac{dy}{dx} = \frac{x^2+y^2}{xy}$ (d) $\frac{dy}{dx} = \frac{xy}{x^2+y^2}$ (e) $\frac{dy}{dx} = x + y$

14. Which of the following is a First Order Exact Ordinary Differential Equation? (a) $(3x^2y + y^2)dx + (x^3 + 2xy)dy = 0$ (b) $(2x^3y)dx + (y^3 + x)dy = 0$ (c) $(ye^x + x)dx + (xe^x + y)dy = 0$ (d) $(2xy^2 + y)dx + (x^2y + x)dy = 0$ (e) $(3x + 2y)dx + (5x + 4y)dy = 0$

15. Which of the following is NOT a First Order Exact Ordinary Differential Equation? (a) $(2xy + y^2)dx + (x^2 + 2xy)dy = 0$ (b) $(xy + 1)dx + (\frac{1}{2}x^2 + y)dy = 0$ (c) $(ye^x + x)dx + (xe^x + y)dy = 0$ (d) $(3x^2y + y^2)dx + (x^3 + 2xy)dy = 0$ (e) $(3x^2y + 2y)dx + (x^3 + 2x)dy = 0$

16. For the following general solution, obtain a particular solution using the given conditions. $y = A\cos 2x + B\sin 2x$, $y(\frac{\pi}{4}) = 3$, and $y'(\frac{\pi}{4}) = 8$. (a) $y = 4\cos 2x + 3\sin 2x$ (b) $y = -4\cos 2x + 3\sin 2x$ (c) $y = -4\cos 2x - 3\sin 2x$ (d) $y = 4\cos 2x - 3\sin 2x$ (e) None of the above

17. Determine the differential equation whose general solution is $y = Ae^{2x} + Be^{-x}$ (a) $y'' + y' - 2y = 0$ (b) $y'' - y' + 2y = 0$ (c) $y'' - y' - 2y = 0$ (d) $y'' + y' + 2y = 0$ (e) None of the above

18. Solve $\frac{dy}{dx} = \frac{x+3y}{2x}$ (a) $(x+y)^2 = Ax^3$ (b) $(1+y)^2 = Ax^3$ (c) $(x+y)^2 = Ax^2$ (d) $(1+y)^2 = Ax^2$ (e) None of the above

19. Solve $x\frac{dy}{dx} - 5y = x^7$ (a) $y = \frac{x^5}{2} + cx^7$ (b) $y = \frac{x^7}{2} + cx^5$ (c) $y = \frac{x^5}{2} - cx^7$ (d) $y = \frac{x^7}{2} - cx^5$ (e) None of the above

20. Solve $x\frac{dy}{dx} - 3y = 4x$ (a) $y = -x + cx^2$ (b) $y = -2x + cx^3$ (c) $y = -\frac{2}{5}x + cx^5$ (d) $y = -5x + cx^2$ (e) None of the above

21. Solve $x\frac{dy}{dx} - 4y = x^3$ (a) $y = Cx^2 - x^3$ (b) $y = Cx^4 + x^3$ (c) $y = Cx^4 - x^3$ (d) $y = Cx^3 - x^4$ (e) None of the above

22. Solve the homogenous differential equation $(x+2y)dx + (2x+3y)dy = 0$ (a) $x^2 + 4xy + 3y^2 = c$ (b) $x^2 - 2xy - 3y^2 = c$ (c) $x^2 - 4xy + y^2 = c$ (d) $x^2 - xy - y^2 = c$ (e) None of the above

23. Solve $y' + y = e^x$ (a) $y = 2e^{-x} + Ce^x$ (b) $y = \frac{1}{2}e^{-x} + Ce^x$ (c) $y = e^x + Ce^{-x}$ (d) $y = \frac{1}{2}e^x + Ce^{-x}$ (e) None of the above

24. Solve $\frac{dy}{dx} + \frac{1}{x}y = xy^2$ (a) $y = (cx - x^2)^{-2}$ (b) $y = (x^2 - cx)^{-2}$ (c) $y = (x^2 - cx)^{-1}$ (d) $y = (cx - x^2)^{-1}$ (e) None of the above

25. Solve $\frac{d^2y}{dt^2} + 2\frac{dy}{dt} - 3y = 0$, $y(0) = 9$, $y'(0) = 5$ (a) $e^{-3t} + 8e^t$ (b) $\beta_1 e^{-3t} + \beta_2 e^t$ (c) $\beta_1 e^{-3t} + \beta_2 e^{-t}$ (d) $e^{-3t} + 8e^{-t}$ (e) None of the above

26. Given that $A\frac{d^2y}{dt^2} + B\frac{dy}{dt} + Cy = 0$, If the discriminant, $B^2 - 4AC > 0$. What is the possible general solution if the roots are m_1 and m_2 ? (a) $\beta_1 e^{m_1 x} + \beta_2 e^{m_2 x}$ (b) $\beta_1 e^{m_1 x} + \beta_2 x e^{m_1 x}$ (c) $e^{m_1 x} (\beta_1 \sin m_1 x + \beta_2 \cos m_1 x)$ (d) $\beta_1 + \beta_2 (e^{-m_1 x} + e^{m_1 x})$ (e) None of the above

27. The second order differential equation $A \frac{d^2y}{dt^2} + B \frac{dy}{dt} + Cy = f(t)$ is said to be homogeneous if (a) $f(t) \neq 0$ (b) $f(t) > 0$ (c) $f(t) = 0$ (d) $f(t) = 0$ or $f(t) > 0$ (e) None of the above ✓
28. Solve $\frac{d^2y}{dt^2} - 2 \frac{dy}{dt} + 5y = 0$, $y(0) = 5$, $y'(0) = 5$ (a) $e^t(5\cos 2t + 2\sin 2t)$ (b) $e^{2t}(5\cos 2t + 2\sin 2t)$ (c) $e^t(5\cos 2t - 2\sin 2t)$ (d) $e^{2t}(5\cos 2t - 2\sin 2t)$ (e) None of the above ✓
29. Solve $\frac{d^2y}{dt^2} + 9 \frac{dy}{dt} + 20y = f(t)$ where $f(t) = 11e^{-4t}$ (a) $11te^{-4t}$ (b) $11te^{-4t} + \beta_1 e^{-5t} + \beta_2 e^{-4t}$ (c) $10e^{3t} + \beta_1 e^{-5t} + \beta_2 e^{-4t}$ (d) $2e^{3t} + \beta_1 e^{-5t} + \beta_2 e^{-4t}$ (e) None of the above ✗
30. The solution to the inhomogeneous part (or the particular solution, y_p) of the differential equation, $\frac{d^2y}{dt^2} + 2 \frac{dy}{dt} - 3y = f(t)$ where $f(t) = \sin 3t$ is (a) $\frac{1}{30}(\sin 3t + 2\cos 3t)$ (b) $-\frac{1}{30}(2\sin 3t - \cos 3t)$ (c) $-\frac{1}{30}(2\sin 3t + \cos 3t)$ (d) $\frac{1}{30}(\sin 3t + 4\cos 3t)$ (e) None of the above ✓
31. A particle moves in a straight line such that for a short time its speed is defined by $v = (3t^2 + 4t)$ m/s where t is in seconds. Determine the distance (s) of the particle when $t = 5$ seconds with initial value of $t = 0$ and $s = 0$ (a) 175m (b) 200m (c) 400m (d) 800m (e) None of the above ✓
32. A particle moves on a straight line from a position $s = 5t^3 + 3t^2$ in metre at time t . Determine the acceleration of the particle at time $t = 3$ sec. (a) 14 ms^{-2} (b) 96 ms^{-2} (c) 100 ms^{-2} (d) 90 ms^{-2} (e) None of the above ✗
33. A car moves in a three-dimensional space such that for a short time its speed is given by $v = (4t + 5, 2t + 3t^3, t + 5t^2) \text{ ms}^{-1}$ where t is the time in seconds. Determine its position in 2 seconds. (a) $\begin{pmatrix} 5 \\ 0 \\ 5 \end{pmatrix} \text{ m}$ (b) $\begin{pmatrix} 3 \\ 1 \\ 4 \end{pmatrix} \text{ m}$ (c) $\begin{pmatrix} 12 \\ 15 \\ 2 \end{pmatrix} \text{ m}$ (d) $\begin{pmatrix} 18 \\ 16 \\ 46 \\ 3 \end{pmatrix} \text{ m}$ (e) None of the above ✗
34. Solve $\frac{d^2y}{dt^2} + 9 \frac{dy}{dt} + 20y = 0$ (a) $\beta_1 e^{-5t} + \beta_2 e^{-4t}$ (b) $\beta_1 e^{-4t} + \beta_2 e^{5t}$ (c) $\beta_1 e^{3t} + \beta_2 e^{6t}$ (d) $\beta_1 e^{4t} + \beta_2 e^{5t}$ (e) None of the above ✓
35. Solve $y'' + 4y' + 3y = 15$ (a) $\beta_1 e^{-2x} + \beta_2 e^{-4x} + 5x$ (b) $\beta_1 e^{2x} + \beta_2 e^{-4x} + 15$ (c) $\beta_1 e^{-2x} + \beta_2 e^{-x} + 5$ (d) $\beta_1 e^{3x} + \beta_2 e^{-x} + 15x$ (e) None of the above ✗